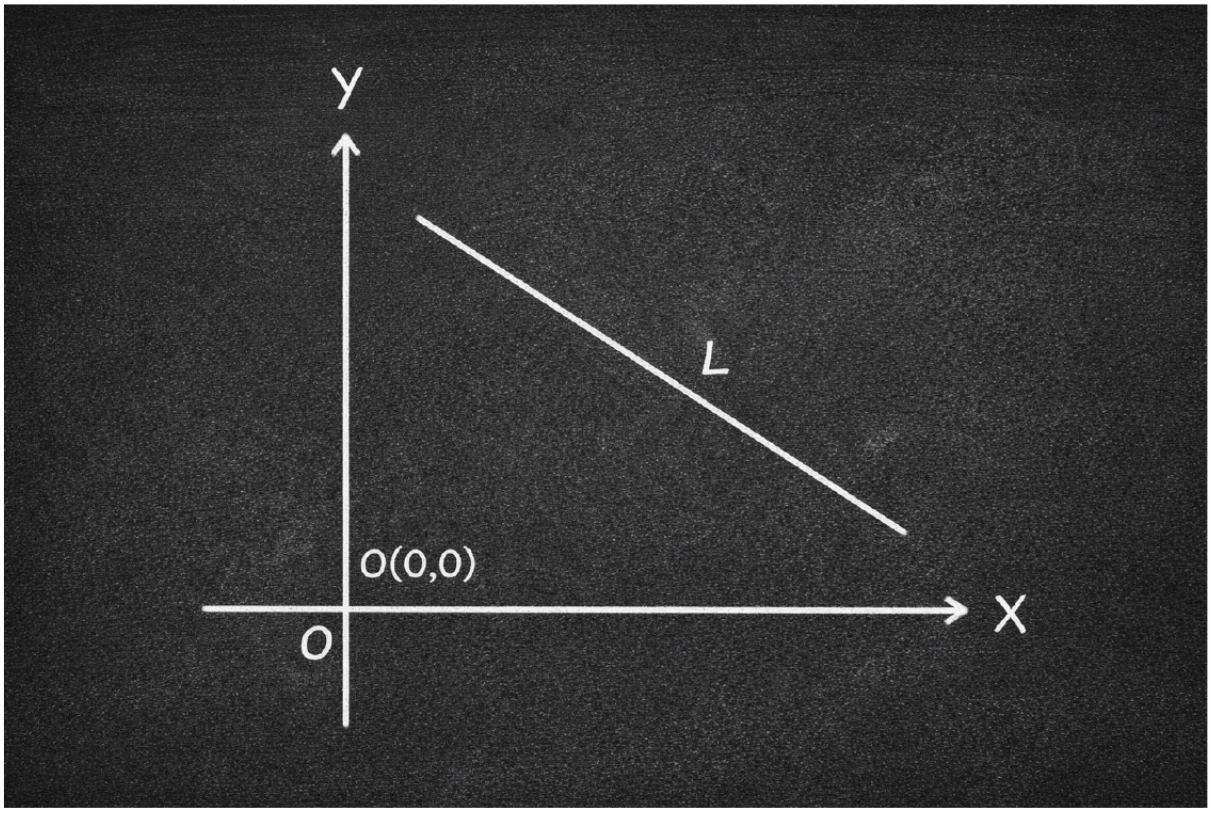
# Unit 3 - Session 2

## SLO 1: Apply the foot-of-normal method for robust line parameterization.

UG Level | Duration: 10 Minutes per Activity

**Question:**




**Q1.** What are the coordinates of the origin marked on the X–Y plane?
Answer: O(0,0).

**Q2.** Does the given line *L*pass through the origin? How can you tell from the diagram?
Answer: No. The line does not pass through the origin because it is separated from O(0,0) in the diagram.

**Q3.** From which point is the perpendicular (normal) drawn to the line *L*?
Answer: The perpendicular (normal) is drawn from the origin O(0,0) to the line.

**Q4.** What is the point where the perpendicular from the origin meets the line *L*called?
Answer: It is called the foot of the normal (foot point).

**Q5.** What does the distance from the origin to this point represent in line parameterization?
Answer: It represents the perpendicular distance ρ from the origin to the line.

**Q6.** What does the angle $\theta$represent in the foot-of-normal method?
Answer: The angle θ represents the angle made by the normal with the positive X-axis.

**Q7.** Write the mathematical expression used to compute $\rho$for a point $(x, y)$on the line.
Answer: ρ = x cos θ + y sin θ.

**Q8.** Why is the foot-of-normal method considered more robust than the slope-intercept form?
Answer: It is more robust because it avoids the undefined slope problem for vertical lines and represents every line using perpendicular distance and angle.